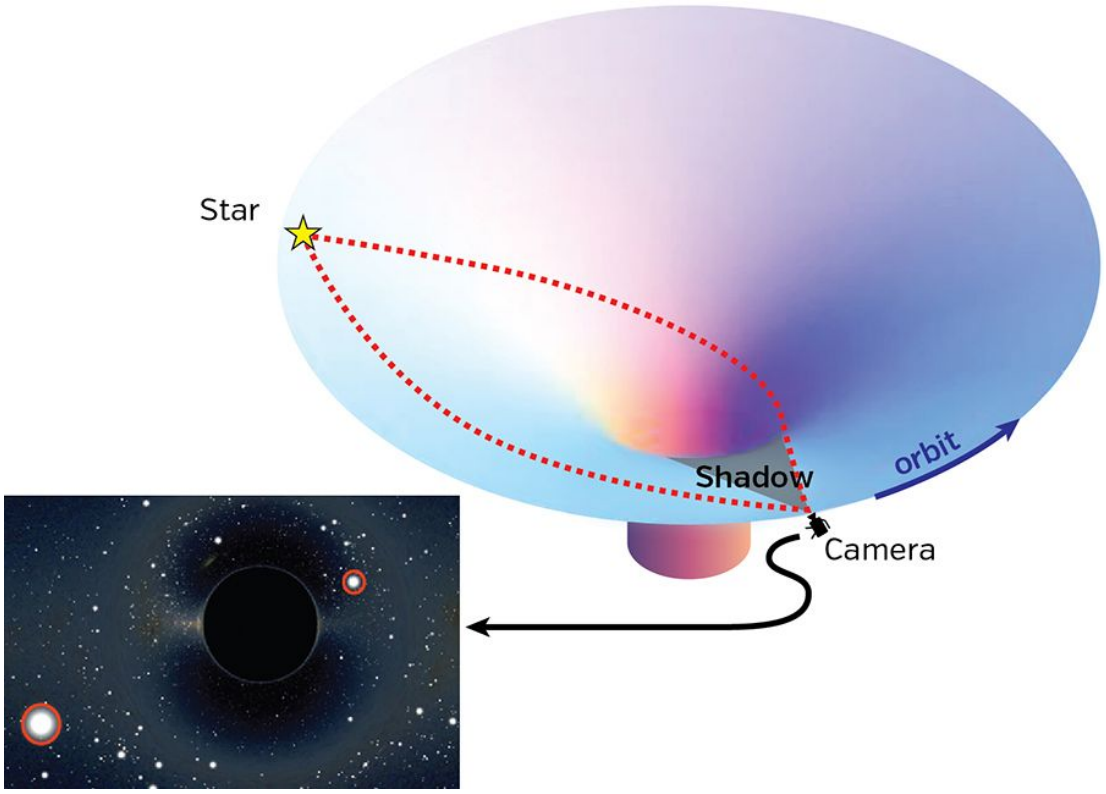


Fig. 8.3. *Top*: The warped space around a nonspinning black hole as seen from the bulk, and two light rays that travel through the warped space from a star to the camera. *Bottom*: The gravitationally lensed pattern of stars that is seen by the camera. *[From a simulation by Alain Riazuelo; for a film clip of his simulation, see www2.iap.fr/users/riazuelo/interstellar .]*

Each of the other stars appears twice in the picture, on opposite sides of the hole's shadow. Can you identify some of the pairs? The black hole's shadow, in the picture, consists of directions from which no rays can come to the camera; see the triangular shaped region labeled "shadow" in the upper diagram. All the rays that "want to be" in the shadow got caught and swallowed by the black hole.

As the camera moves rightward in its orbit (Figure 8.3), the pattern of stars seen by the camera changes as shown in Figure 8.4.

This figure highlights two particular stars. One is circled in red (the same star circled in Figure 8.3). The other is inside a yellow diamond. We see two images of each star: one image is outside the pink circle; the